Syllabus Details

Syllabus ID:	13822
Syllabus Name:	**Mobile Programming_Lập trình di động**
Course Name English:	**Mobile Programming**
Subject Code:	**PRM393**
Learning-Teaching Method:	Project based learning, In-class lecture, Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 104h self-study
Pre-Requisite:	PRO192
Description:	1. Understand the fundamental concepts of mobile programming with Flutter and Dart. 2. Experience using popular widgets and plugins to build cross-platform user interfaces. 3. Knowledge of state management, form handling, and navigation between screens in mobile applications. 4. Learn and apply asynchronous programming techniques with Future and async/await. 5. Ability to integrate RESTful APIs to exchange data between mobile applications and servers, as well as store data locally using SharedPreferences and SQLite. 6. Proficient in using Dart libraries and command-line tools to optimize the application development process. 7. Knowledge of implementing mobile notification systems, including local and push notifications.
StudentTasks:	- Students must attend more than 80% of contact sessions in order to be accepted to the final examination. - Student is responsible to do all assigned exercises given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the FU LMS at http://lms.fpt.edu.vn for up-to-date course information
Tools:	- Android Studio + Flutter SDK + Dart SDK (Latest Version) - Virtual device attached android studio (or other virtual device), android mobile (only for learning, not for exam - optional) - Laptop: 16GB RAM, 25GB free space before installing studio Note: - Install all neccessary tools before the first slot of the course
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1402/QĐ-ĐHFPT dated 12/12/2025
IsApproved:	**True**
Note:	- Visual Studio Code (recommended, lightweight and supports many Flutter/Dart extensions). - Xcode (only needed if building & testing iOS, required on macOS).
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	12/12/2025

4 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Mastering Flutter: Learn to Develop Flutter Apps for iOS, Android, Desktop and Web	Kevin Moore	BPB Publications	2025	1st	9789365899177	True	True	False	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
2	Beginning App Development with Flutter: Create Cross-Platform Mobile Apps	Rap Payne	Apress	2019	1st	9781484251805	False	False	True	
3	Developing Mobile Apps with Flutter Specialization	IBM Skills Network Team	Coursera				False	False	True	https://www.coursera.org/specializations/de mobile-apps-with-flutter
4	E-learning Platforms						False	False	True	https://flutter.dev https://docs.flutter.dev https://dart.dev/guides/language/language- https://github.com/flutter/samples

7 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand the fundamental concepts of mobile programming with Flutter and Dart.
2	CLO2	Experience using popular widgets and plugins to build cross-platform user interfaces.
3	CLO3	Knowledge of state management, form handling, and navigation between screens in mobile applications.
4	CLO4	Learn and apply asynchronous programming techniques with Future and async/await.
5	CLO5	Ability to integrate RESTful APIs to exchange data between mobile applications and servers, as well as store data locally using SharedPreferences and SQLite.
6	CLO6	Proficient in using Dart libraries and command-line tools to optimize the application development process.
7	CLO7	Knowledge of implementing mobile notification systems, including local and push notifications.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Flutter Overview & Environment Setup	Offline	CLO1	ITU	M1 – Introduction to Flutter. Reference: Ch.1 Getting Started, tooling, project structure.	PRM393	Verify setups; troubleshoot common install issues; walk-through run configurations.	
2	Project Structure & Tooling	Offline	CLO1, CLO6	TU	M1 – Introduction to Flutter. Reference: Ch.1 Getting Started, tooling, project structure		Explain folder structure; demonstrate build flavors; answer Q&A.	
3	Hot Reload/Restart & Widget Tree	Offline	CLO1, CLO6	TU	M1 – Introduction to Flutter. Reference: Ch.1 Getting Started, tooling, project structure.		Demo hot reload/hot restart; show DevTools widget inspector.	
4	Basic UI Composition with Scaffold	Offline	CLO1, CLO6	TU	M1 – Introduction to Flutter. Reference: Ch.1 Getting Started, tooling, project structure.		Live-code scaffold customization; review students' changes.	
5	Summary & Setup Issues	Offline	CLO1, CLO6	TU	M1 – Introduction to Flutter. Reference: Ch.1 Getting Started, tooling, project structure.		Summarize pitfalls; collect environment reports; set expectations.	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
6	Dart Types & Variables	Offline	CLO1, CLO4	TU	M2 – Dart Essentials. Reference: Ch.2–3 Dart syntax, types, functions, classes.		Review type system & null safety; address common errors.	
7	Functions & Functional Patterns	Offline	CLO1	TU	M2 – Dart Essentials. Reference: Ch.2–3 Dart syntax, types, functions, classes.		Demonstrate function patterns; code review live.	
8	Classes & Constructors	Offline	CLO1, CLO2	TU	M2 – Dart Essentials. Reference: Ch.2–3 Dart syntax, types, functions, classes.		Explain class design; refactor student solutions.	
9	Collections & Operations	Offline	CLO1, CLO2	TU	M2 – Dart Essentials. Reference: Ch.2–3 Dart syntax, types, functions, classes.		Show idiomatic collection usage; performance tips.	
10	Intro to Futures & Async	Offline	CLO2, CLO3	TU	M2 – Dart Essentials. Reference: Ch.2–3 Dart syntax, types, functions, classes.		Introduce async model; clarify event loop concepts.	
11	Future-based Async Programming	Offline	CLO1, CLO3	TU	M3 – OOP & Asynchronous. Reference: Ch.3–4 OOP patterns, Futures, Streams, async/await.		Explain Future scheduling; demonstrate error propagation.	
12	Streams & StreamController	Offline	CLO1, CLO3	TU	M3 – OOP & Asynchronous. Reference: Ch.3–4 OOP patterns, Futures, Streams, async/await.		Show StreamController patterns; memory-safety notes.	
13	StreamBuilder in Flutter	Offline	CLO1	TU	M3 – OOP & Asynchronous. Reference: Ch.3–4 OOP patterns, Futures, Streams, async/await.		Live-code StreamBuilder; discuss rebuild behavior.	
14	Repository Pattern	Offline	CLO1, CLO2	TU	M3 – OOP & Asynchronous. Reference: Ch.3–4 OOP patterns, Futures, Streams, async/await.		Present repository abstraction; layering guidance.	
15	Reactive Patterns (Concept)	Offline	CLO1, CLO2	TU	M3 – OOP & Asynchronous. Reference: Ch.3–4 OOP patterns, Futures, Streams, async/await.		Discuss reactive patterns; pitfalls and best practices.	
16	Progress Test 1	Offline	CLO1, CLO2, CLO3	U	Progress Test 1		Progress Test 1	
17	UI Basics – Scaffold Layout ListView & Card/ListTile	Offline	CLO1	TU	M4 – UI Fundamentals. Reference: Ch.4 Widgets, Layouts, MaterialApp, lists & pickers.		Demonstrate composition; spacing best practices. Code-along list patterns; performance considerations.	
18	Interactive Widgets	Offline	CLO1, CLO3	TU	M4 – UI Fundamentals. Reference: Ch.4 Widgets, Layouts, MaterialApp, lists & pickers.		Handle onChanged; explain ephemeral vs app state.	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
19	Pickers – Date/Time	Offline	CLO3	TU	M4 – UI Fundamentals. Reference: Ch.4 Widgets, Layouts, MaterialApp, lists & pickers.		Show pickers & result handling; UX guidance.	
20	Theming & UI Polishing	Offline	CLO3	TU	M4 – UI Fundamentals. Reference: Ch.4 Widgets, Layouts, MaterialApp, lists & pickers.		Review theming knobs; show common polish tips.	
21	Navigator 1.0 – Push/Pop	Offline	CLO1	TU	M5 – Navigation & State Management. Reference: Ch.8 Navigator 1.0/2.0, named routes, deep links, basic state. Session focus: Part 1.		Explain Navigator stack; demo simple flow.	
22	Named Routes & Arguments	Offline	CLO1, CLO6	TU	M5 – Navigation & State Management. Reference: Ch.8 Navigator 1.0/2.0, named routes, deep links, basic state.		Show RouteSettings & argument handling patterns.	
23	Navigator 2.0 Concepts	Offline	CLO1, CLO7	TU	M5 – Navigation & State Management. Reference: Ch.8 Navigator 1.0/2.0, named routes, deep links, basic state.		Present RouterDelegate/RouteInformationParser overview.	
24	Android Deep Linking	Offline	CLO1, CLO5	TU	M5 – Navigation & State Management. Reference: Ch.8 Navigator 1.0/2.0, named routes, deep links, basic state.		Walk-through manifest intent-filters; test live.	
25	iOS URL Scheme Navigation	Offline	CLO1	TU	M5 – Navigation & State Management. Reference: Ch.8 Navigator 1.0/2.0, named routes, deep links, basic state.		Guide Info.plist & associated domains; QA routing. Assign and confirm project topics	
26	Flex Layout (Row/Column)	Offline	CLO1, CLO3	TU	M6 – Layout & Responsive Design. Reference: Ch.5–6 Advanced layouts (Grid/Stack/Flex), responsive patterns.		Explain constraints; demonstrate Flex factors.	
27	GridView & Stack	Offline	CLO1, CLO3	TU	M6 – Layout & Responsive Design. Reference: Ch.5–6 Advanced layouts (Grid/Stack/Flex), responsive patterns.		Show grid/stack patterns; layering tips.	
28	LayoutBuilder & MediaQuery	Offline	CLO5	TU	M6 – Layout & Responsive Design. Reference: Ch.5–6 Advanced layouts (Grid/Stack/Flex), responsive patterns.		Introduce breakpoints; compute layout decisions.	
29	Responsive UI Design	Offline	CLO7, CLO2	TU	M6 – Layout & Responsive Design. Reference: Ch.5–6 Advanced layouts (Grid/Stack/Flex), responsive patterns.		Critique responsive solutions; UX considerations.	
30	Responsive Refactor	Offline	CLO1, CLO5	TU	M6 – Layout & Responsive Design. Reference: Ch.5–6 Advanced layouts (Grid/Stack/Flex), responsive patterns.		Refactor example; code review and feedback.	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
31	Progress Test 2	Offline	CLO3, CLO4, CLO5	U	Progress Test 2		Progress Test 2	
32	Forms & Controllers Validation & Errors	Offline	CLO1	TU	M7 – Forms & Validation. Reference: Chapter 4 (Basic Widgets) & Chapter 14 (User Input) Chapter 4 (Form Widgets) & Chapter 14 (Validation UX)		Explain controllers lifecycle; disposal rules. Demonstrate validation patterns; UX for errors.	
33	Submit Flow & Data	Offline	CLO1	TU	M7 – Forms & Validation. Reference: Chapter 14 (Input Flow), Chapter 6 (State Concepts)		Show submit flow; data handling & confirmation.	
34	Form Draft Save	Offline	CLO1	TU	M7 – Forms & Validation. Reference: Chapter 6 (State Basics) & Chapter 14 (Input Events)		Discuss draft-saving approaches; data model notes.	
35	Focus & Keyboard Actions	Offline	CLO1, CLO7	TU	M7 – Forms & Validation. Reference: Chapter 14 (Focus, Keyboard Actions)		Show focus traversal; accessibility considerations.	
36	HTTP Request & JSON Parsing	Offline	CLO2, CLO7	TU	M8 – Networking & JSON API. Reference: Ch.9 HTTP/REST, JSON parsing, FutureBuilder.		Explain HTTP basics; JSON to model mapping.	
37	FutureBuilder & List Rendering	Offline	CLO1, CLO2	TU	M8 – Networking & JSON API. Reference: Ch.9 HTTP/REST, JSON parsing, FutureBuilder.		Live-code FutureBuilder; list rendering tips.	
38	Error/Loading UI	Offline	CLO1, CLO2	TU	M8 – Networking & JSON API. Reference: Ch.9 HTTP/REST, JSON parsing, FutureBuilder.		Error/empty/loading design; UX standards.	
39	Detail Screen via API ID	Offline	CLO1, CLO6	TU	M8 – Networking & JSON API. Reference: Ch.9 HTTP/REST, JSON parsing, FutureBuilder.		Route to detail by ID; data flow sanity checks.	
40	Retry & Caching	Offline	CLO1	TU	M8 – Networking & JSON API. Reference: Ch.9 HTTP/REST, JSON parsing, FutureBuilde		Discuss caching layers; avoiding redundant calls.	
41	SharedPreferences Basics	Offline	CLO1	TU	M9 – Local Storage & Persistence. Reference: Ch.10 SharedPreferences, SQLite/Hive, caching..		Contrast prefs vs database; data lifecycle.	
42	Local DB (SQLite/Hive)	Offline	CLO1	TU	M9 – Local Storage & Persistence. Reference: Ch.10 SharedPreferences, SQLite/Hive, caching.		Schema modeling; CRUD operations overview.	
43	Persistent List Rendering	Offline	CLO3, CLO5	TU	M9 – Local Storage & Persistence. Reference: Ch.10 SharedPreferences, SQLite/Hive, caching.		Render persisted data; integrity checks.	
44	Sync-Up Concept	Offline	CLO1	TU	M9 – Local Storage & Persistence. Reference: Ch.10 SharedPreferences, SQLite/Hive, caching.		Discuss sync patterns; conflict resolution basics.	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
45	Export/Backup JSON	Offline	CLO1	TU	M9 – Local Storage & Persistence. Reference: Ch.10 SharedPreferences, SQLite/Hive, caching.		Show export concept; data portability discussion.	
46	Login Form & Mock Auth	Offline	LO1, CLO2	TU	M10 – Authentication & User Management. Reference: Ch.11 Auth basics, session, guarded routes.		Explain auth concepts; security caveats.	
47	Session State Management	Offline	CLO3	TU	M10 – Authentication & User Management. Reference: Ch.11 Auth basics, session, guarded routes.		Session/state handling; update UI accordingly.	
48	Guarded Routes	Offline	CLO2	TU	M10 – Authentication & User Management. Reference: Ch.11 Auth basics, session, guarded routes.		Route guarding patterns; navigation decisions.	
49	Token Persistence (Mock)	Offline	CLO1	TU	M10 – Authentication & User Management. Reference: Ch.11 Auth basics, session, guarded routes.		Persist token mock; discuss security limits.	
50	Integrated Auth Flow	Offline	CLO1	TU	M10 – Authentication & User Management. Reference: Ch.11 Auth basics, session, guarded routes. Session focus: Part 5.		End-to-end demo; troubleshooting common issues.	
51	Unit Tests for Project Features	Offline	LO1, CLO2	TU	M11 – Testing & Debugging. Core concepts from Ch.12.		Explain testing setup in project context; assist integration.	
52	Widget Tests & Debugging	Offline	CLO7	TU	M11 – Testing & Debugging. Core debugging tools.		Review test coverage; show debugging techniques.	
53	Progress Test 3	Offline	CLO5, CLO6, CLO7	U	Progress Test 3		Progress Test 3	
54	Sprint 3 Review Refactoring & Polishing	Offline	CLO1, CLO6	TU	Project Work – Sprint 3 Review. Project Work – Polishing & Refactoring.		Conduct sprint review; give improvement feedback. Provide code review session; guide refactoring priorities.	
55	User Testing & Report	Offline	CLO1	TU	Project Work – User Testing.		Supervise test execution; monitor report documentation.	
56	Build & Release Config	Offline	CLO1	TU	M12 – Build & Release. Ch.13 (condensed).		Demonstrate build process (debug/release).	
57	Packaging & Release Build	Offline	CLO1	TU	Project Work – Packaging.		Guide deployment process; validate final artifacts.	
58	Presentation Rehearsal	Offline	CLO1	TU	Project Work – Presentation Rehearsal.		Mentor on presentation skills; storytelling & flow.	
59	Mock Defense	Offline	CLO1	TU	Final Evaluation – Mock Defense.		Act as evaluator; simulate defense Q&A.	
60	Final Project Submission	Offline	CLO1	U	Final Project Presentation		Grade presentation; summarize learning outcomes.	

0 Constructive question(s)

4 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Practical Exam	on-going	1	25.0%	4	90'/each	All CLOs	Write a program	By instructor, in class	studied chapters knowledge and skills	Students prepare personal computer with Android studio and virtual machines installed. The exam proctor will monitor and grade exam directly in the exam room.	1 × 25% =25% of to progress mark (each student wr a small program)
2	Progress Test	on-going	3	15.0%	>0	Option 1: 30'; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	PT1: CLO1, CLO2, CLO3; PT2: CLO3, CLO4, CLO5; PT3: CLO5 CLO6,CLO7	Option 1: Multi-choice Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1:25; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	studied chapters knowledge and skills	by instructor	3 × 5% = 15% of tota progress mark
3	Project	on-going	1	30.0%	>0	Option 1: At home; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	All CLOs	Option 1: N/A; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	Option 1: N/A; Option 2 (For Constructivism Approach only): Follow lecturer's proposal	All chapters; knowledge and skills of mobile programming	by instructor	(20% product, 10 presentatio each stude chooses project by himself
4	Final exam	Final exam	1	30.0%	4	60'	All CLOs	Multi-Choice	50	All chapters; knowledge and skills of mobile programming	by exam board	